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RESEARCH ARTICLE

Effects of task-oriented training on dexterous movements of hands in post stroke patients

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ABSTRACT

Objective: The objective of the study was to determine the effect of task-oriented training on the dexterous movements of hands in Hemiplegic post-stroke patients.
Methods: This study has been registered on site ClinicalTrial.gov with clinical trial number NCT05201196. The study was Randomized controlled trial, 18 patients were recruited that meet the inclusion criteria, randomly allocated to task-oriented training Group A ($n=9$) and Conventional Therapy Group B ($n=9$). Both exercise trainings were applied for 45 min/session, 5 times/week for 6 weeks. Fugl-Meyer Assessment Scale Motor, sensory and coordination portion, Wolf Motor Function Scale and Barthel Index were used as outcome measures, assessed patients at Baseline, after 3 weeks and 6 weeks after training. Data were analyzed by SPSS version 23.
Results: The results suggested the mean Age was 60.78 ± 9.08 and 61.33 ± 6.78 for Group A and Group B, respectively. Average BMI was 23.66 ± 2.66 for Task-oriented group and 21.36 ± 2.46 for Conventional group. Fugl-Meyer scale shows significant P-value 0.03 post treatment compare to pre-treatment which was .283, Wolf Motor Function test and Barthel Index also showed significant *P*-values as 0.023 and 0.007, respectively, indicating that Task-oriented training shows more significant improvements than conventional group.
Conclusion: Task-based training produced statistically significant as well as clinically meaningful enhancement in the dexterous hand movements of acute and subacute stroke patients than conventional therapy and ultimately improves the functional independence in their daily activities such as feeding, bathing and hygiene.

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Stroke; Rehabilitation; hand movements; task performance; motor performance; neuroplasticity

Introduction

Cerebrovascular accident (CVA), also known as stroke, is a neural condition that primarily disturbs the arteries leading to and inside the brain. It results when the blood supply to that part of the brain is diminished and disrupted. When the brain cells die during a stroke, it causes a sudden loss of consciousness and paralysis [1]. Neurological abnormalities such as sensory, motor, cognitive, linguistic and emotional difficulties are common in stroke patients. Decreased self-efficacy, psychological and emotional disturbances (such as anxiety and depression) are all symptoms of these illnesses in stroke patients, which lower the quality of life [2]. Impaired motor performance is related to damage to the corticospinal tract (CST) on the ipsilesional side [3]. CST is related to the fine motor skills of the distal part of the extremities, especially with the activities of the hand, as CST controls the activation of hand muscles [4]. According to the recent studies, when there is damage to CST, there is an up regulation of the reticulospinal tract which results in the flexor synergies of arm and finger enslavement on the paralytic side and the mirror movements on the non-paralytic side. This increased regulation of the reticulospinal tract is also seen in chronic stroke patients with minor to moderate paresis during the recovery period [5]. Damage to upper motor neurons is characterized by both positive and negative manifestations. The positive manifestations include tendon jerk, hypertonia, released flexor reflexes (Babinski reflex), clonus and extensor stretch reflex; while paresis, loss of dexterity, loss of fine motor control, hypotonia, and increased muscle

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fatigability are the negative ones that are shown in the early phases of neuron damage [6]. In post-stroke patients, arm paralysis appears to be a bi-model manifestation. Most patients suffering from stroke have either severe arm paralysis where the use of their arms is limited in performing activities of daily life, or mild-to-moderate arm paralysis in which they can function with the paretic hand but still lack dexterity. The ability to perform quick, precise wrist and finger motions (wrist-finger speed), operate small objects (finger dexterity), and manipulate larger ones (manual dexterity), our capacity to keep our arm still (steadiness), move it rapidly and accurately toward a goal (aiming), or move it along a plane under continual visual control (tracking), all of these abilities are deficits in stroke patients with slight to moderate arm paralysis [7]. Voluntary movements tend to recover first at the shoulder, then the elbow, then the wrist and fingers in stroke patients. These suggest that recovery of the movements of the upper limb occurs in a proximal to distal direction [8]. The healing of the upper extremity is slower than that of the lower extremity, making rehabilitation more difficult [9]. Deterioration in hand functions occurs in the chronic stage of stroke, which is significantly apparent as loss of dexterity (negative sign) as well as decreased finger strength, and abnormal flexion synergy of the hand, identified by the flexion of the hand and fingers (positive signs) due to involuntary motor activation pattern. Unintentional force production by uninformed fingers and increased finger enslavement are the most eminent deficits in hand dexterity [10]. Motor cortex and corticospinal tract lesions are considered mainly responsible for abnormal hand functions, as they control the independent movements of fingers [11]. Recovery of finger strength and finger individuation either occur independently or relative to each other [12]. The Principle of Neuroplasticity provides the therapeutic basis for stroke patient rehabilitation. The process in the brain that results in the formation of new neurons and connections between these neurons as a consequence of learning is called neuroplasticity. This occurs by executing repetitive and dynamic tasks with a paralyzed upper extremity [13]. The gold standard for post-stroke rehabilitation is task-specific training based on motor learning principles, which is used in both contemporary treatment techniques and clinical research [14]. Task-specific approach should be initiated as soon as possible after the injury with main emphasis on relearning movements as it is based on the motor relearning principles [15]. Different terms are interchangeably used for this approach such as task-specific training, task-oriented training, repetitive task training or task-related training [16]. The notion of plasticity and motor learning in rehabilitation is reflected in the fact that recurrent practice may help the integration of residual altered sensory and motor systems [17]. Task-specific training entails the active practice of motor tasks within a defined functional setting, which may include complicated whole-limb or pre-task motions of the entire limb or a limb segment [18]. A perfect task-oriented protocol is one that involves both unilateral and bilateral activities. These can be performed repeatedly individually or in groups [19]. The present study helps to recover the functional independence in acute and chronic Stroke patients by improving fine movements of their hand, which play a crucial part in performing daily activities.

Methodology

In this Study, 18 patient were recruited that meet the inclusion criteria, with unilateral hemiplegia referred by Neurophysician to the Rehabilitation Department of Tehsil Headquarter Hospital (THQ) Khushab for Rehabilitation from October 2021 to March 2021, having stroke for the first time, between aged 40 and 70 year, both males and females, score of spasticity for upper extremity (shoulder, elbow) below and equal 2 based on the Modified Ashworth Scale (MAS), ability to comprehend simple instructions (Mini-Mental State Examination with a minimum score > 24), Brunnstrom stages ≥ 4 were included and the individuals with recurrent stroke episodes and transient ischemic attack, other neurological diseases (Parkinson's disease, multiple sclerosis,), Hemineglect, no sitting balance and co-morbidities were excluded. Outcome measures used were Fugl-Meyer Assessment Scale (wrist and hand) and the Wolf motor function test for upper extremity motor impairment and ability assessment, while Barthel index was use to assess the activities of daily living.

Treatment protocol

These outcome measures were admisterated at initial at Baseline(pre-treatment), middle at the third week and post-treatment at the sixth week [20]. A total of 18 post-stroke hemiplegic patients who fulfil the inclusion and exclusion criteria were randomly allocated by lottery method into two groups, nine participants in each, Study was explained to the participants and informed consent taken from all the participants. The interventions were applied in sitting position.

Group A: Task-oriented group

Group A received intervention was task-oriented training therapy for 5 days/week for 6 weeks period with each session was performed for 45 min. Task-oriented training was based on the previous research include (Table-top polishing, Arm cradling, Reach forward and pick-up or touch an object, Reached sideway to pick-up an object and transferring it to a table in front, Pouring ½ cup of water from a measuring pot into wide mouth glass held in opposite hand, Picking up pen from thumb and first two fingers, lifting a basket and placing it on the table) (Figure 1) [16]

Group B: Conventional therapy group

Group B received conventional treatment (self-devised including stretching, strengthening exercises, TENS and heating). Transcutaneous electrical nerve stimulation and heating was applied for 10-15 min, targeted muscle stretching and strengthening (Isometrics) of flexors, extensors, abductors, internal and external rotators, of shoulder, elbow, forearm (supination and pronation) and hand 10 repetitions × 1 set, 5 days/week. Total of 16 sessions were given each consisting of 45 min/day for 6 weeks (Figure 2).

Data collection procedure

Approval obtained from the ethical committee prior to the commencement of study. The pre-and post-evaluation of hand functions and activities of daily living was performed using the Fugl-Meyer Assessment Scale for upper extremity, Wolf motor function test for upper extremity and Barthel Index for evaluating quality of life.

Fugl-Meyer upper extremity assessment (FMA-UE)

Fugl-Meyer upper extremity evaluates voluntary movement, reflex activity, coordination, and grasp. Performance was measured on with a 3-point ordinal scale (0 to 2), with a maximum score of 30 with sub-score 10 for the wrist, 14 for the hand, and for coordination and a speed of movement the score was 6. The score for pain and Passive Range of Motion assessment was 24 for each component.

Wolf motor function test (WMFT)

Wolf motor function is a quantitatifiable evaluation of the upper limb motor cability including timed and functional tasks. It comprises of 15 function-based

Figure 1. Task-oriented training including; table top polishing, arm cradling and pen holding.

Figure 2. Conventional therapy; stretching and strengthening exercises.

tasks and 2-strength-based tasks. Maximum score is 75 and lower scores are suggests low level of functioning. Time obligatory for the task vary with individuals and usually 15–20 min.

Barthel index

Barthel index include the data gained from the patient's self report, or from one of his attendents. It include 10 activities of daily living (transferring from chair/wheelchair to bed and back,feeding, bathing, personal hygiene, dressing, controlling bowl and bladder control, getting on and off toilet, walking on level surface/propelling wheelchair, ascending and descending stairs) [16].

Randomization

The randomization process was accomplished using computer-generated randomization method.

Blindness

The assessor who collected post treatment data were blinded by not informing about the group the patient was assigned to.

Data analysis

After collecting data, Shapiro–Wilk test was applied which suggest that data were normally distributed so, parametric tests Repeated Measure ANOVA and One Way ANOVA were used for within-group and between-group analysis, respectively. For analysis, SPSS version 23 was used.

Descriptive analysis was performed for demographic characteristics of patients which were described in terms of Mean ± Standard Deviation, frequencies and percentages.

Results

Of all Patients participated in study have completed the study period. The mean age and standard deviation of patients was 60.78 ± 9.08 for task-specific group; while 61.33 ± 6.78 for conventional group. The mean of BMI was 23.66 ± 2.66 and 21.36 ± 2.46 for task-oriented and conventional group, respectively (Table 1). The study includes 13 (72.22%) females and 5 (27.78%) males, out of which 6(66.67%) females and 3 (33.33%) males were included in the task-oriented group and 2 (22.22%) males and 7 (77.78%) females were included in the conventional group. Almost 90% patients were affected from ischemic stroke and only 10% patients were suffering from hemorrhagic stroke, the distribution of these baseline demographic characteristics in both treatment groups in terms of frequencies and percentages are mentioned in detail in Table 1.

Table 1. Demographic and clinical characteristics of participants.

Characteristics	Task-oriented training group	Conventional therapy group
	(*n* = 9)	**(*n* = 9)**
Age (years)	60.78 ± 9.08	61.33 ± 6.782
BMI	23.66 ± 2.66	21.36 ± 2.466
Gender		
Male	3 (33.33%)	2 (22.22%)
Female	6 (66.67%)	7 (77.78%)
Hemisphere		
Right	4 (44.44%)	6 (66.67%)
Left	5 (55.56%)	3 (33.33%)
Side of hemiplegia		
Right	5 (55.56%)	3 (33.33%)
Left	4(44.44%)	6(66.67%)
CVA Types		
Ischemic	7 (77.78%)	9 (100%)
Hemorrhagic	2 (22.22%)	0 (0.00%)
Dominant hand		
Right	7 (77.78%)	8 (88.89%)
Left	2 (22.22%)	1 (11.11%)
Comorbid conditions		
Hypertension	5 (27.78%)	4 (22.22%)
Diabetes mellitus	0 (0.00%)	1 (5.56%)
Both HTN + DM	2 (11.11%)	4 (22.22%)
Others	2 (11.11%)	0 (0.00%)

Notes: Values are expressed in Mean ± Standard Deviation, and numbers (%).
BMI, Body Mass Index; HTN, hypertension and DM, Diabetes mellitus.

In the current study, the *P*-value was.260 and .907 for task-oriented training and conventional therapy groups, respectively, which are above than $p \geq 0.05$ mean that present study met the assumption of sphericity. *F*-value was 96.809, which reaches significance with a *P*-value of .000 for task-oriented group and for conventional group the *F*-value was 100.0 with a *P*-value of .000 (which is less than 0.05 alpha levels). This mean there was a statistically significant difference between means of different levels of within the subject's variable. Wilk's lambda value was zero (.000) in results of the current study which mean that null hypothesis was rejected and alternate hypothesis was accepted.

One Way ANOVA was used for between group analyses. The *F*-value determines the *P*-value in ANOVA. The results show statistically significant difference between groups for all FMA-UE, WMFT and Barthel Index. In the present study results *P*-value of

Fugl-Meyer was significant at 6 weeks post-treatment (0.03) as compared to pre-value (0.283); this value was below $p \le 0.05$ shows significant results for Fugl-Meyer. *P*-value for Wolf motor function test was also significant post-treatment 0.023 as compare to pre-treatment which was .666. The *P*-value shows significant results for WMFT as it was $p \le 0.05$. Additionally, Barthel Index Score was significantly improve post-treatment as *p*-value was 0.013 at 3 weeks post-treatment and .007 at 6 weeks post-treatment (Table 2).

Comparison of FMA-UE score improvements of both groups (before and after treatment)

(Figure 3).

Table 2. One-way ANOVA for between-group comparisons.

Variable	Task-oriented training group **Mean ± SD (Mean ± SD)**	Conventional therapy group **(Mean Mean ± SD ±SD)**	Between-group *P*-value	*F*-value
FMA UE pre-treatment	47.22 ± 14.57	40.00 ± 12.97	.283	1.233
FMA UE 3 weeks	71.22 ± 7.64	63.33 ± 9.53	.115	2.778
FMA UE 6 weeks	84.22 ± 4.71	79.33 ± 7.43	.03	3.748
WMFT pre-treatment	22.11 ± 9.71	19.77 ± 12.58	.666	.194
WMFT 3 weeks	37.33 ± 6.80	30.33 ± 11.59	.138	2.440
WMFT 6 weeks	50.88 ± 7.07	39.55 ± 11.54	.023	6.304
Barthel Index pre-treatment	28.88 ± 15.36	27.22 ± 10.9	.794	.070
Barthel Index 3 weeks	59.44 ± 8.81	43.88 ± 14.09	.013	7.879
Barthel Index 6 weeks	81.67 ± 7.9	66.11 ± 12.69	.007	9.739

Note: Values of Mean ± Standard Deviation; FMA-UE, Fugl-Meyer Assessment Upper Extremity; WMFT, Wolf motor function test; *F*-value, *P*-value.

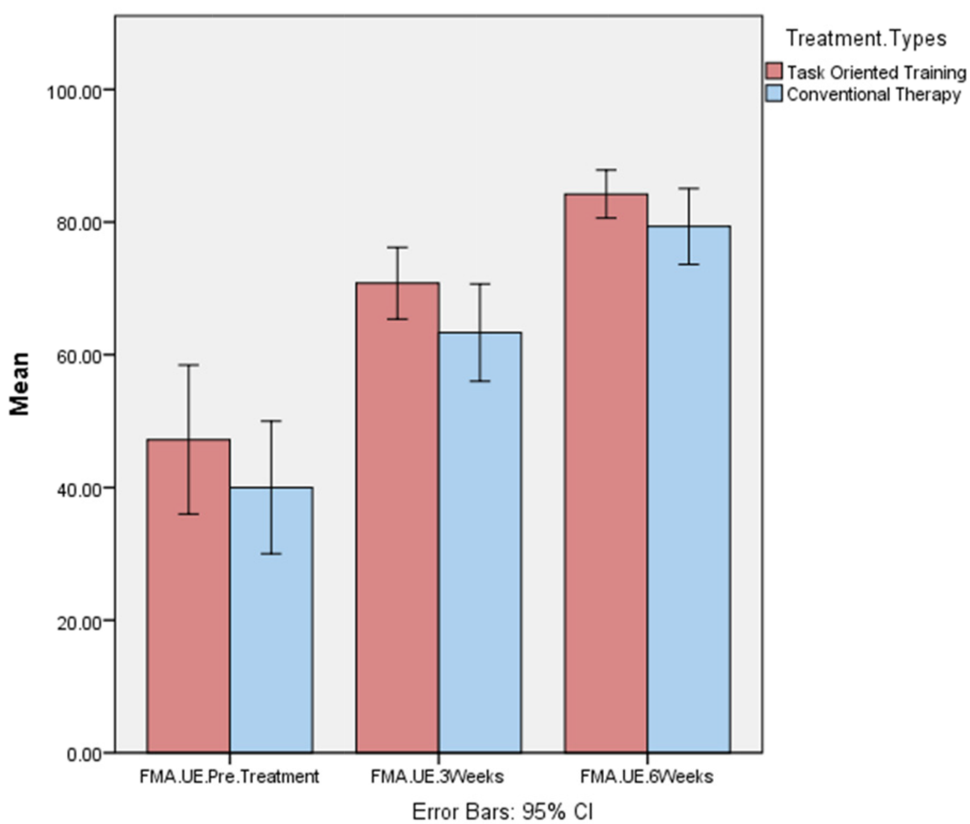

Figure 3. The above mentioned bar chart depict that both groups (task-oriented and conventional) receiving their respective interventions shows improvements on FMA-UE. However, task-oriented training show more significant improvement, and the confidence interval was 95%.

Line chart comparison of both groups (WMFT)

(Figure 4).

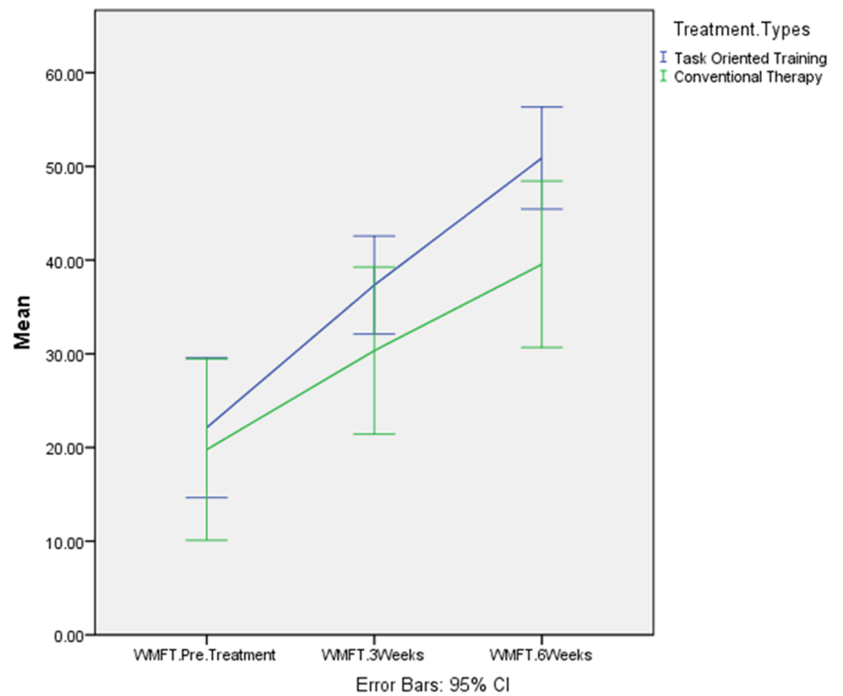

Figure 4. This above-mentioned linear graph represents the effectiveness of task-oriented group and conventional group with time at the *x*-axis and mean at the *y*-axis, according to the Wolf Motor Function score. Wolf motor fuction score is significantly improve for task-oriented group.

Line chart comparison of both groups (Barthel Index)

(Figure 5).

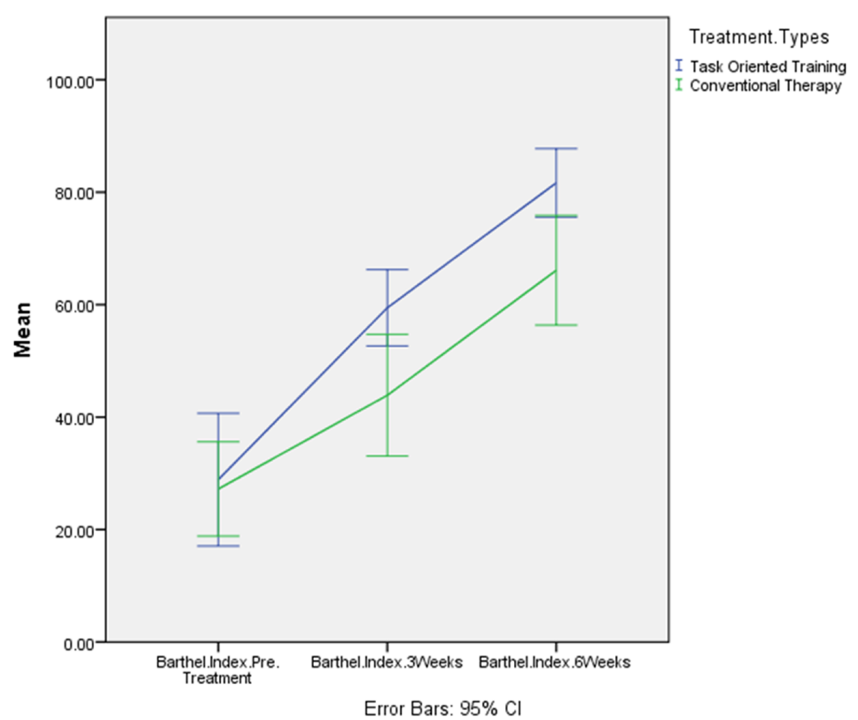

Figure 5. This linear graph indicate the functional improvement measured by Barthel Index Score for Both groups suggest that task-oriented training provide more significant results despite of the fact that both groups show to improve functional performance.

Discussion

This present, randomized controlled trial was conducted to estimate the improvements in dexterous movements of hand in CVA patients by executing task-oriented and conventional therapy for 6 weeks (30 sessions), including 18 stroke patients (both male and females) with 9 patients were randomly assigned in each group. The outcomes suggested that both groups show improvement in fine movements of hand but task-oriented training show more significant improvements confirmed by post-treatment score of FMA-UE, WMFT and Barthel Index. FMA-UE shows improvements in grip strength, pain, coordination and speed of movement; while improvements on the WMFT Score were shown by improved functional ability and grip strength. Greater improvement in ADLs and overall quality of life was shown by Barthel Index score.

The study conducted by Chen et al. [21] to evaluate the effect of robot aided therapy and task-specific training in chronic stroke patients. The results suggested that both therapy approaches improve an upper limb function but task-specific therapy indicate more improvement in body function and robot-assisted training improve the activity domain. The task-based training greatly enhance the body function and spontaneous use of upper limb confirmed by FMA-UE, while the activity performance also showed positive results proved by WMFT and Motor Activity Log which were parallel with the current study in which TOT improve the FMA-UE as well as WMFT score [21].

Previously, study conducted on the suitability of home-based task-related training for arm function improvement in subacute stroke patients, in which Wanner et al. [22] used video-supported task-based training protocol. Study suggested that home program was feasible and the focus on the fact that repetitive task training promote motor learning and motor performance. The outcome of this study was parallel with the current study as repetitive functional training improved the movement of hand [22].

Randomized controlled trial was conducted by Kim and Kim [23] in Korea shows that task-based training was effective in large movements of the upper extremity and hand function when apply with percussion instruments while the control group also show improvements in the upper extremity and hand strength. The outcome of the above-mentioned study was contrast with the current study results as there was no significant difference between groups [23].

In post-stroke patients, the rehabilitation needs to focus on the improvement of ADLs to enhance functional independence. Song et al. [13] conducted an interventional study on the chronic stroke patients in Korea; the results of this study showed significant increase in the function of the affected upper extremity after intervention in both task-based and non-task-based training groups. However, there was no statistical significance between the two groups in the group comparison, which was conflicted with the present study as the current study shown in significant improvement [13].

A research by Alwhaibi et al. [24] compare the outcome of the upper extremity repetitive task training on excitability of brain of the involved hemisphere and improved motor function in patients with the right and left hemisphere damage. They found that repetitive task training produced a substantial improvement in the functional ability of the affected upper limb, and confirmed by post-treatment scores of WMFT and BBT, which supporting the results of the current study [24].

The previous study conducted in Myanmar by Thant et al. [25] proposed that 20 h (4 week) of task-specific training improved motor regaining, functional performance of the upper limb, and quality of hand function more than conventional treatment in subacute stroke patients, which indicated by the decreased time score of WMFT and improved functional ability and grip strength score of WMFT, showed similarity with the recent study whom results suggested an enhanced grip strength confirmed by WMFT score [25].

An experimental study conducted by Badawy [26] on the improving hand function in subacute stroke patients using task-specific training in the Cairo University in Egypt. Both groups receiving physiotherapy program includes PNF, reaching in different directions and weight bearing and was executed for 6 weeks. In addition group one was give task-specific training (eating with spoon, drinking with cup, hair combing, brushing teeth and wearing shirt) for 30 min. The results revealed that task-specific training provide more improvement by functionally improving daily life activities and hand functions shows similarity with the present study [26].

Repetitive functional training improves physical activity in stroke patients. Martins et al. [27] conducted research in Brazil to assess the improvements in physical activity in stroke patient executing repetitive task training for 12 weeks, the control group involve memory training, stretching and education about health. The outcome suggests that task training was more suitable, relevant and beneficial; hence, provide enhanced improvements in physical activity and a mobility level, which were similar to the outcomes of the current study that also improves physical activity and independence in daily activities corroborate by the improved score of the Barthel Index [27].

Conclusion

Task-based training produced statistically significant as well as clinically meaningful enhancement in the dexterous hand movements of acute and subacute stroke patients than conventional therapy and ultimately improves the functional independence in their daily activities such as feeding, bathing and hygiene.

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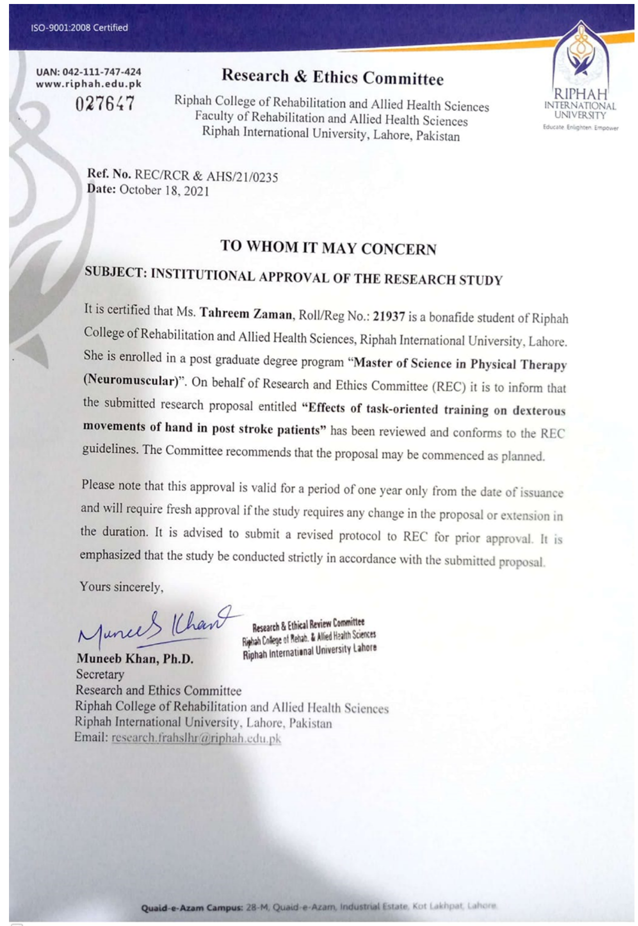
Disclosure statement

No potential conflict of interest was reported by the author(s).

Ethical committee form



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Ref. No. REC/RCR & AHS/21/0235
Date: October 18, 2021

TO WHOM IT MAY CONCERN

SUBJECT: INSTITUTIONAL APPROVAL OF THE RESEARCH STUDY

It is certified that Ms. **Tahreem Zaman**, Roll/Reg No.: **21937** is a bonafide student of Riphah College of Rehabilitation and Allied Health Sciences, Riphah International University, Lahore. She is enrolled in a post graduate degree program **"Master of Science in Physical Therapy (Neuromuscular)"**. On behalf of Research and Ethics Committee (REC) it is to inform that the submitted research proposal entitled **"Effects of task-oriented training on dexterous movements of hand in post stroke patients"** has been reviewed and conforms to the REC guidelines. The Committee recommends that the proposal may be commenced as planned.

Please note that this approval is valid for a period of one year only from the date of issuance and will require fresh approval if the study requires any change in the proposal or extension in the duration. It is advised to submit a revised protocol to REC for prior approval. It is emphasized that the study be conducted strictly in accordance with the submitted proposal.

Yours sincerely,

Muneeb Khan

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Ethical approval

The Riphah college of Rehabilitation and Allied Health Sciences Ethical committee approved this study. All participants gave written informed consent before data collection initiated.

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References

[1] Kim S-H, Won K-A, Jung E-H. The effects of simultaneous application of peripheral nerve sensory stimulation

and task-oriented training to improve upper extremity motor function after stroke: single blinded randomized controlled trial. Therapeutic Sci Rehabil. 2020;9(4):7–20.
[2] Sung Y-B, Lee D-H, Lee J-H. Descriptive study on rehabilitation treatment methods for improving upper limb function in stroke patients. IJANER. 2020;5(2):19–28.
[3] Stinear CM, Byblow WD, Ackerley SJ, et al. PREP2: a biomarker-based algorithm for predicting upper limb function after stroke. Ann Clin Transl Neurol. 2017;4(11):811–820.
[4] Karadag A, Mirkhasilova M, Middlebrooks EH, et al. Importance of different characteristic of the corticospinal tract based on DTI and cadaveric microdissection. J Basic Clin Health Sci. 2021 May 1;5(2):65–72.
[5] Hammerbeck U, Tyson SF, Samraj P, et al. The strength of the corticospinal tract not the reticulospinal tract determines upper-limb impairment level and capacity for skill-acquisition in the Sub-acute post-stroke period. Neurorehabil Neural Repair. 2021;35(9):812–822.
[6] Patel TS, Kanase S. Effect of task oriented training on spasticity in post stroke individuals. JEOH. 2021;21(3):90–95.
[7] Platz T, Lotze M. Arm ability training (AAT) promotes dexterity recovery after a stroke—a review of its design, clinical effectiveness, and the neurobiology of the actions. Front Neurol. 2018;9:1082.
[8] Schambra HM, Xu J, Branscheidt M, et al. Differential poststroke motor recovery in an arm versus hand muscle in the absence of motor evoked potentials. Neurorehabil Neural Repair. 2019;33(7):568–580.
[9] Adhikari SP, Tretriluxana J, Chaiyawat P, et al. Enhanced upper extremity functions with a single session of action-observation-execution and accelerated skill acquisition program in subacute stroke. Stroke Res Treat. 2018;2018:1490692.
[10] Xu J, Ejaz N, Hertler B, et al. Separable systems for recovery of finger strength and control after stroke. J Neurophysiol. 2017;118(2):1151–1163.
[11] Mawase F, Cherry-Allen K, Xu J, et al. Pushing the rehabilitation boundaries: hand motor impairment can be reduced in chronic stroke. Neurorehabil Neural Repair. 2020;34(8):733–745.
[12] Wolbrecht ET, Rowe JB, Chan V, et al. Finger strength, individuation, and their interaction: relationship to hand function and corticospinal tract injury after stroke. Clin Neurophysiol. 2018;129(4):797–808.
[13] Song B-K, Han S-Y, Jeong J-W, et al. The effect of task-based upper limb training on activities of daily living and upper limb function in chronic stroke patients. Medico Legal Update. 2020;20(1):2069–2073.
[14] Krakauer JW, Cortés JC. A non-task-oriented approach based on high-dose playful movement exploration for rehabilitation of the upper limb early after stroke: a proposal. NeuroRehabilitation. 2018;43(1):31–40.
[15] Kuberan P, Kumar V, Joshua AM, et al. Effects of task oriented exercises with altered sensory input on balance and functional mobility in chronic stroke: a pilot randomized controlled trial. Bangladesh J Med Sci. 2017;16(2):307–313.
[16] Firoozeh F, Noorizadeh S, Dadgoo M, et al. A comparison among task oriented training with and without bobath program on upper limb in stroke patients. Funct Disab J. 2019;2(1):83–90.
[17] Divya S. Influence of number of repetition and variety of task in upper limb motor recovery in hemiplegia. J Phys Med Rehabil Stud Reports SRC/JPMRS. 2020;/127:3.
[18] da Silva ESM, Ocamoto GN, Santos-Maia G, et al. The effect of priming on outcomes of Task-Oriented training for the upper extremity in chronic stroke: a systematic review and Meta-analysis. Neurorehabil Neural Repair. 2020;34(6):479–504.
[19] Martins JC, Nadeau S, Aguiar LT, et al. Efficacy of task-specific circuit training on physical activity levels and mobility of stroke patients: a randomized controlled trial. NRE. 2020;47(4):451–412. (Preprint):
[20] Adeagbo CA, Olawale OA, Gbiri CAO. Transcranial direct current stimulation and repetitive functional task-oriented programme for upper limb functional rehabilitation in stroke survivors. Phys Ther Rev. 2021;26(6):420–427.
[21] Chen YW, Chiang WC, Chang CL, et al. Comparative effects of EMG-driven robot-assisted therapy versus task-oriented training on motor and daily function in patients with stroke: a randomized cross-over trial. J Neuroeng Rehabil. 2022;19(1):6.
[22] Wanner M, Schönherr G, Kiechl S, et al. Feasibility of an individualised, task-oriented, video-supported home exercise programme for arm function in patients in the subacute phase after stroke: protocol of a randomised controlled pilot study. BMJ Open. 2022;12(1): e051504.
[23] Kim J-H, Kim M-K. Task-oriented approach using percussion instruments in chronic stroke patients effect on upper limb function: a randomized controlled trial. KSPM. 2021;16(4):103–116.
[24] Alwhaibi RM, Mahmoud NF, Zakaria HM, et al. A comparative study on the effect of task specific training on right versus left chronic stroke patients. IJERPH. 2020;17(21):7950.
[25] Thant AA, Wanpen S, Nualnetr N, et al. Effects of task-oriented training on upper extremity functional performance in patients with Sub-acute stroke: a randomized controlled trial. J Phys Ther Sci. 2019;31(1):82–87.
[26] Badawy WM. Effect of task-oriented exercises on improving hand function in subacute stroke patients: a randomized controlled trial. Environment. 2017;4:5.
[27] Martins JC, Aguiar LT, Nadeau S, et al. Efficacy of task-specific training on physical activity levels of people with stroke: protocol for a randomized controlled trial. Phys Ther. 2017;97(6):640–648.